LocalPoint3D · Mid-Project Report

Track 3: ML Model and Dataset Development | Group 2

Can 3D recognition models trained on Western datasets recognise Bangladeshi household objects captured with commercial smartphone scanning?

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2 November 2026 · Code, data tables and detailed reports: github.com/Md-Mosabbir/localpoint3d

> **In one sentence:** PointLLM, a 3D model trained on Western data, recognises our everyday scans (15 of 17 correct) but **none of our 14 Bangladeshi object scans (0 of 14)**. It still reads their shape, but always picks the nearest familiar object instead: rehal → bench, patil → bowl, bodna → vase/teapot. A CLIP-style model makes the same kind of mistake.

1. Problem statement (updated)

Problem. Most 3D recognition models are trained on Western datasets: ModelNet40, ShapeNet and Objaverse. Objects common in Bangladesh, such as the bodna, rehal, kula and piri, or vehicles like the rickshaw and CNG, hardly appear in them. We test whether these models still recognise such objects when we capture them ourselves with **commercial smartphone scanning** (the Scaniverse app). We also build the small, cleaned dataset of local objects that this test needs.

What changed once we started, and why

Initial plan	Now	Why
Compare iPhone LiDAR and photogrammetry as two capture methods	**Commercial smartphone scanning** (Scaniverse) as one capture method; the LiDAR/photo comparison is a side question	Our first results showed the biggest problem was **clutter** in the scan (floor, walls), not the capture mode. The literature review on LiDAR vs photogrammetry is done; our own measurement is deferred.
Start with vehicles (rickshaw, thelagari, CNG)	Start with **household objects** (bodna, piri, kula, muura, patil, rehal, matir cup, hurricane lamp); vehicles later	Small objects are fast to scan and clean, so we could fix the method quickly before scanning big objects outdoors.
Use raw scans	**Hand-clean every scan** (CloudCompare) + automatic speck removal	Raw scans with floor and walls were described as “scenes” (“a ranch”, “a beach scene”). Cleaning raised correct answers from 24 to 33 of 51.
Ask open questions (“What is this?”)	Add a **multiple-choice test** with “blind” inputs (a ball, random noise)	Open answers are hard to score and vary run to run. Multiple choice plus blind inputs shows whether the model uses the **shape** or just guesses common words.
Colour scans	**Grey** scans (shape only) for the main test	We want to know whether the **shape** is recognised. On clean scans colour changed little (33 vs 32 of 51).
PointLLM-V2 as the main model	**PointLLM v1.2** + a CLIP-style model	PointLLM-V2 weights are not released (checked 21 Oct). Our supervisor asked for a CLIP comparison too.

2. What we finished so far

2.1 Capture and dataset: 34 cleaned scans

We scanned objects with Scaniverse on iPhones, exported point clouds (PLY), cleaned each scan by hand in CloudCompare and stored them on Google Drive with a fixed naming scheme (*Object_take_cleaned.ply*).

Set	Objects	Scans
Everyday (reference)	chair 8, cup 4, keyboard 3, table 2	17
Bangladeshi	piri 4, bodna 2, kula 2, patil 2, matir cup 1, muura 1, rehal 1, hurricane lamp 1, wooden bench 3	17

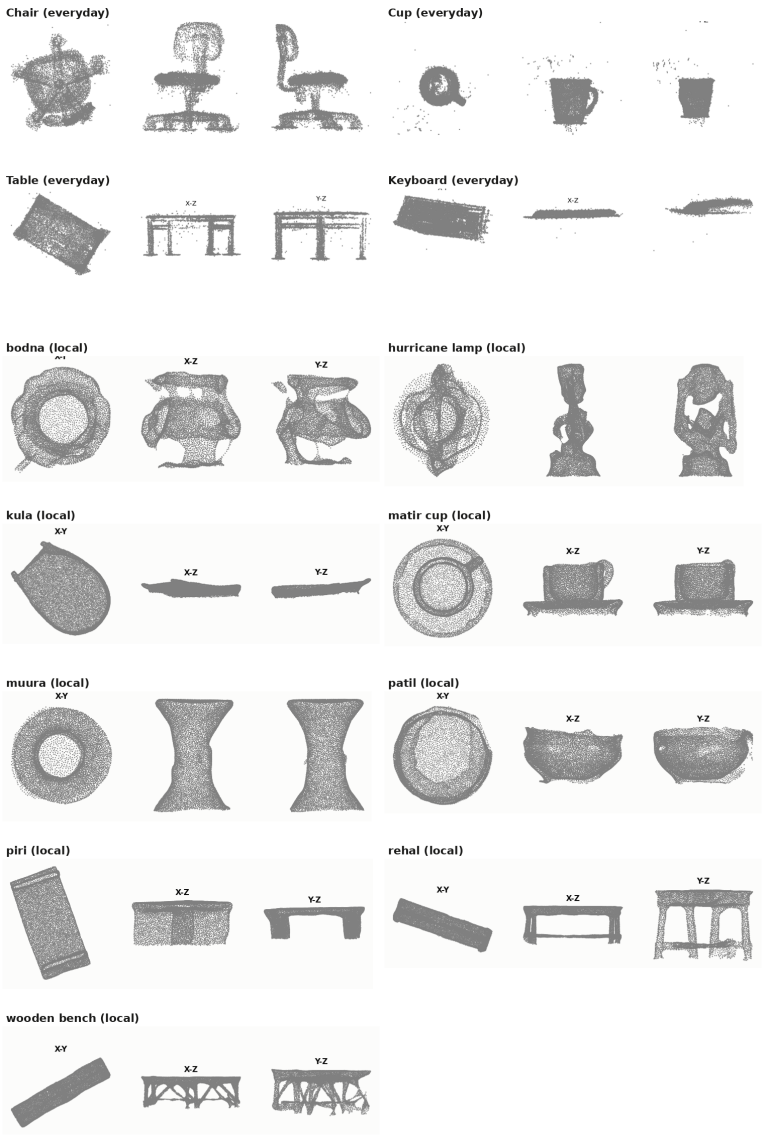


Figure 1. What the model sees: each scan reduced to 8192 points, centred, scaled and greyed. Three views per scan (top, front, side).

What it contributes: the dataset itself (Track 3) and a capture-and-cleaning protocol that every later test relies on.

2.2 A reproducible test pipeline

- **Pipeline:** phone scan → hand cleaning → Google Drive → Google Colab (free T4 GPU) → model → CSV → analysis. Three Colab notebooks, with shared Python code in the repository.

- **Pre-processing:** 8192 points, unit sphere, optional grey, automatic floor and speck removal. A single stray speck far away used to shrink the object to a dot.
- **Repeatable results:** PointLLM turned out to be random even with fixed settings (its point encoder starts from a random point). We fixed the random seed: two runs now give **identical** scores to the last decimal.

What it contributes: every result in this report can be re-run by any member, and the same pipeline will test vehicles and other models.

2.3 Everyday objects: baseline and the effect of cleaning

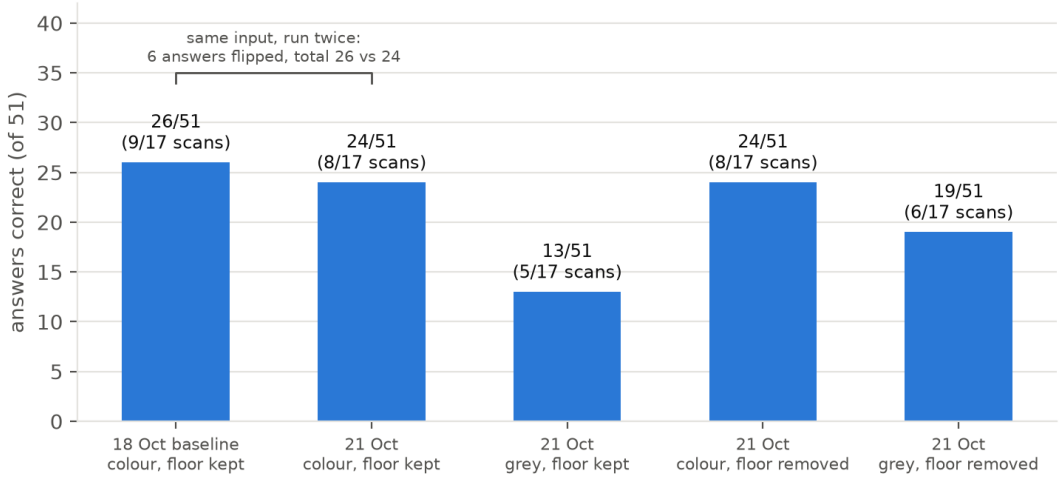


Figure 2. Raw scans, 18 and 21 Oct: answers correct of 51. Without colour (grey) the score halved; removing the floor automatically helped grey scans only. The same input run twice flipped 6 answers: PointLLM was random before we fixed the seed.

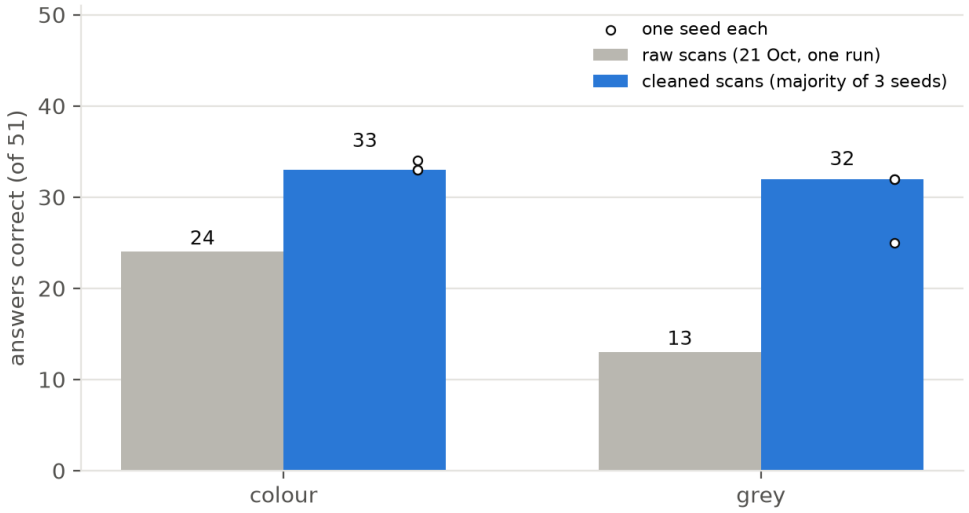


Figure 3. PointLLM, 17 everyday scans, 3 questions each (51 answers). Cleaning the scans helped most, especially in grey. Dots: the three seeds.

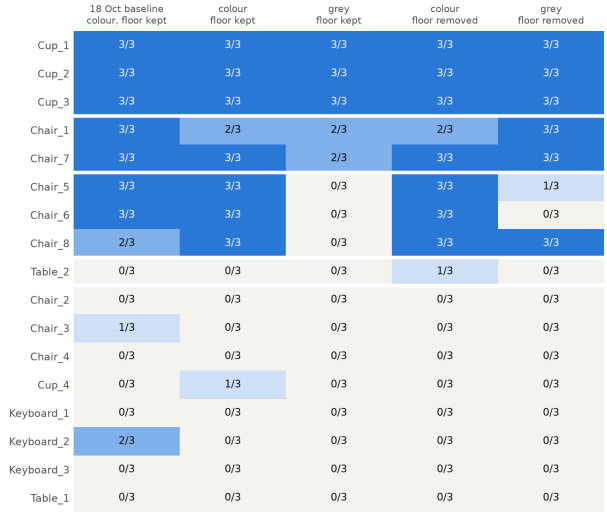


Figure 4. Raw scans, every scan (correct of 3 questions). Cups and most chairs work; tables, keyboards and Chairs 2–4 fail in every setting.

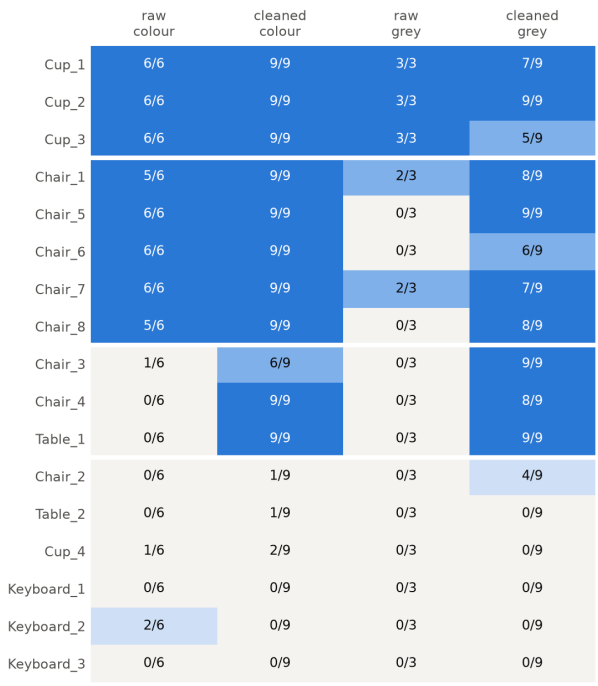


Figure 5. Raw vs hand-cleaned, every scan. Cleaning fixed Chair_03, Chair_04 and Table_1 and made most grey chairs work; keyboards, Table_2, Cup_4 and Chair_02 still fail.

Run	Result
18 Oct, raw scans	9 of 17 scans named correctly
21 Oct, raw, colour vs grey	24 vs 13 of 51 answers correct: colour seemed essential
23 Oct, **hand-cleaned**	**33** (colour) and **32** (grey) of 51; 11 of 17 scans recognised. Once clean, colour hardly matters

What it contributes: proof that the pipeline works on objects the model knows. Without this baseline, a failure on local objects could just mean a bad scan.

2.4 Bangladeshi objects, open questions (25 Oct)

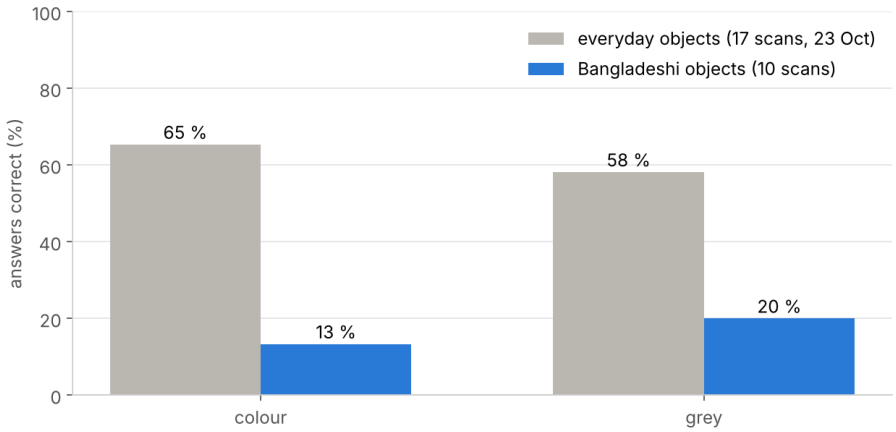


Figure 6. Share of answers correct, same settings. Everyday 65 % / 58 %, Bangladeshi 13 % / 20 % (colour / grey).

	colour	grey	what PointLLM usually said
Matir_Cup_1	9/9	9/9	cup and saucer
Piri_2	1/9	5/9	chair / pillow / rock
Bodna_1	0/9	3/9	Pokémon figure; grey: pot
Muura_1	2/9	1/9	cartoon bird
Rehal_1	0/9	0/9	bench (legs at 45°)
Kula_1	0/9	0/9	flat board / stone
Kula_2	0/9	0/9	flat board / broken plate
Piri_3	0/9	0/9	rectangular block
Piri_4	0/9	0/9	block / rock / table
Hurricane_Lamp_1	0/9	0/9	sculpture of a woman with a child

Figure 7. The 10 local scans of 25 Oct, correct answers of 9 (3 questions × 3 seeds), and what PointLLM usually said. Correct = the nearest English word (e.g. pot for bodna, stool or chair for muura).

Only the matir cup was recognised (“a cup and saucer”). The model often described the **shape** correctly but named the wrong object: rehal → “a bench with legs at 45 degrees” (18 of 18), kula → “a flat, oval-shaped board”. When unsure it invented familiar things (hurricane lamp → “a sculpture of a woman”, bodna → “a Pokémon character”).

What it contributes: the first direct answer to the research question: the local objects are mostly not recognised.

2.5 Are the local objects “near out-of-distribution”?

Near-OOD means a new kind of object that looks like something the model knows. We checked this in two ways.

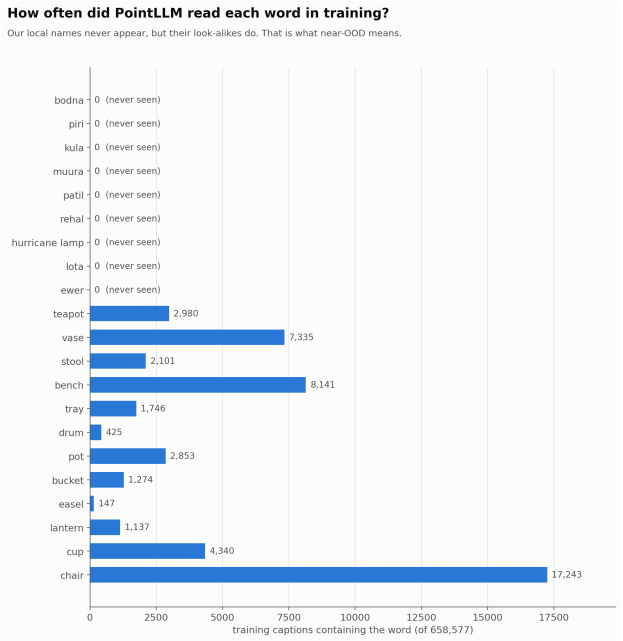


Figure 8. Number of PointLLM's 658,577 training captions that contain each word. The local names never appear; their look-alikes are common.

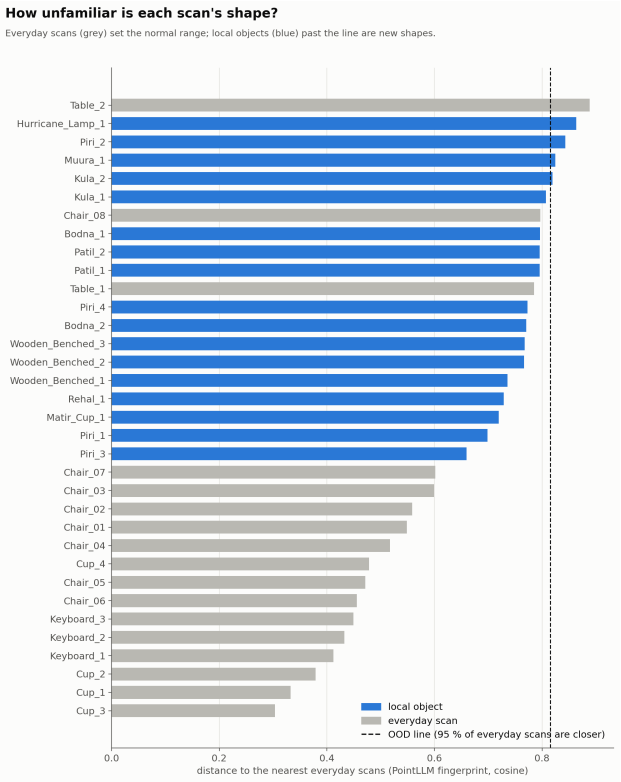


Figure 9. How far each scan is from the everyday scans in PointLLM's own "shape fingerprint". Local scans (blue, 0.66–0.86) are farther than almost all everyday scans (mostly 0.30–0.60).

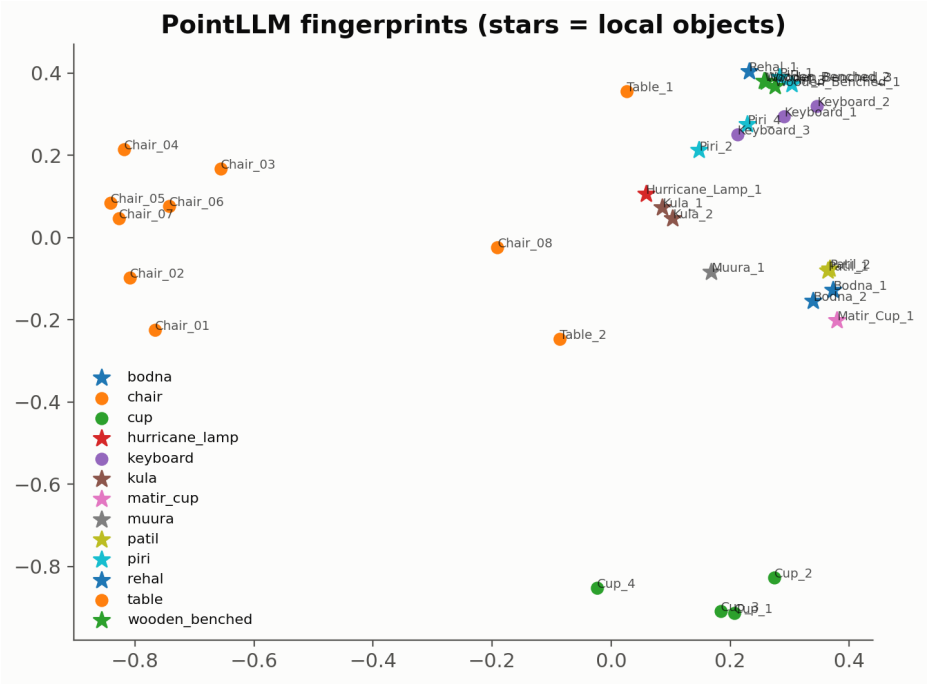


Figure 10. PointLLM's shape fingerprints flattened to 2D (dots = everyday, stars = local). Local objects form their own groups, next to their look-alikes: piri and rehal near the wooden benches and keyboards, patil, bodna and matir cup together, away from the cups.

What it contributes: evidence that our objects are genuinely new to the model, both by name and by shape.

2.6 Multiple choice in grey: does it read the shape? (31 Oct)

For each scan PointLLM chose between 4 answers: the correct one, a look-alike, a related object and an unrelated object (25 % by guessing). It does not write anything: we measure how likely it finds each answer. We also asked the same question about a featureless ball, random noise and a different object (“blind” inputs).

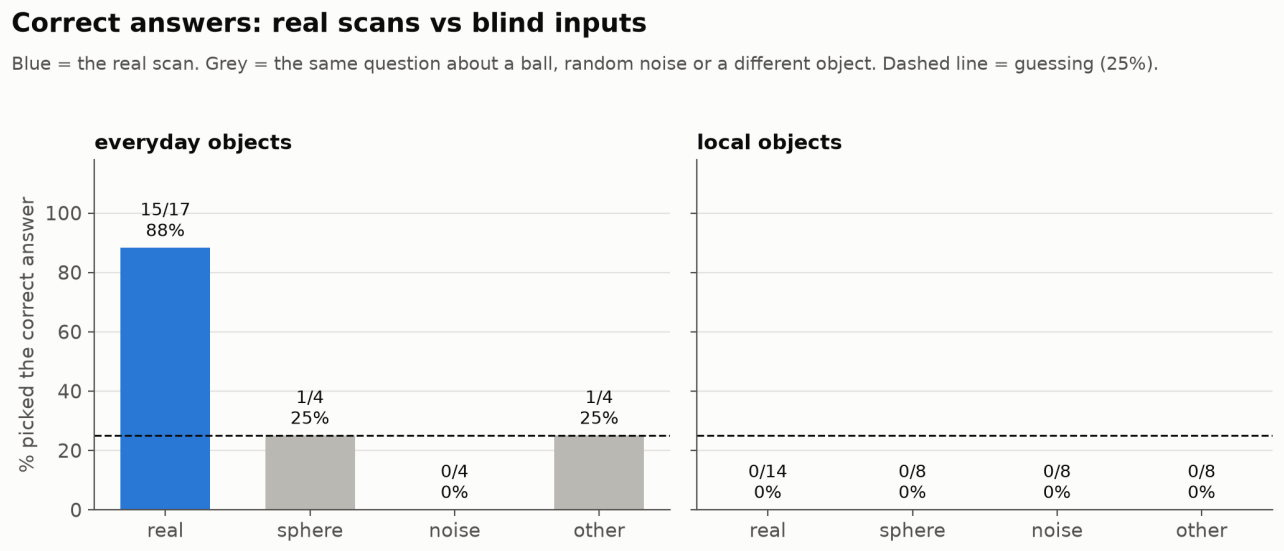


Figure 11. Everyday scans 15 of 17 correct, while the blind inputs are at chance. Bangladeshi scans 0 of 14, in two runs with differently worded answers.

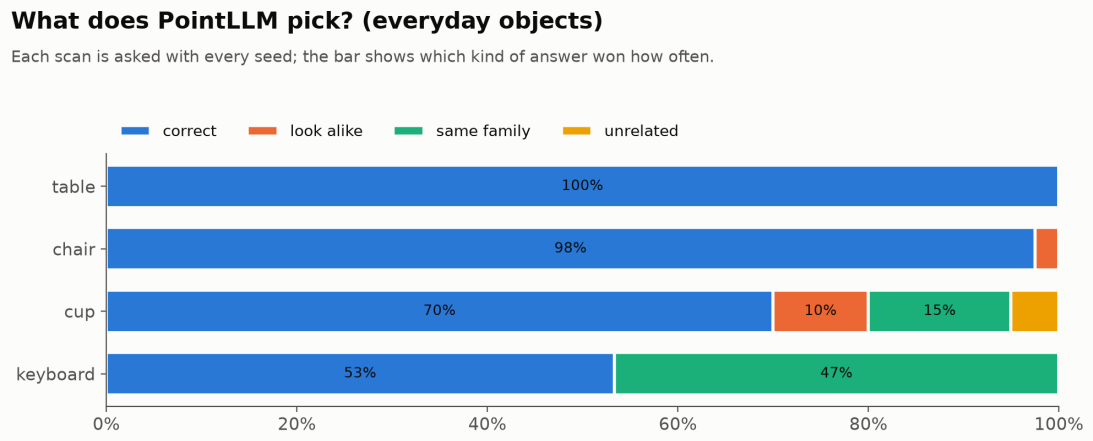


Figure 12. Everyday objects: the correct answer wins most of the time (tables 100 %, chairs 98 %, cups 70 %, keyboards 53 %).

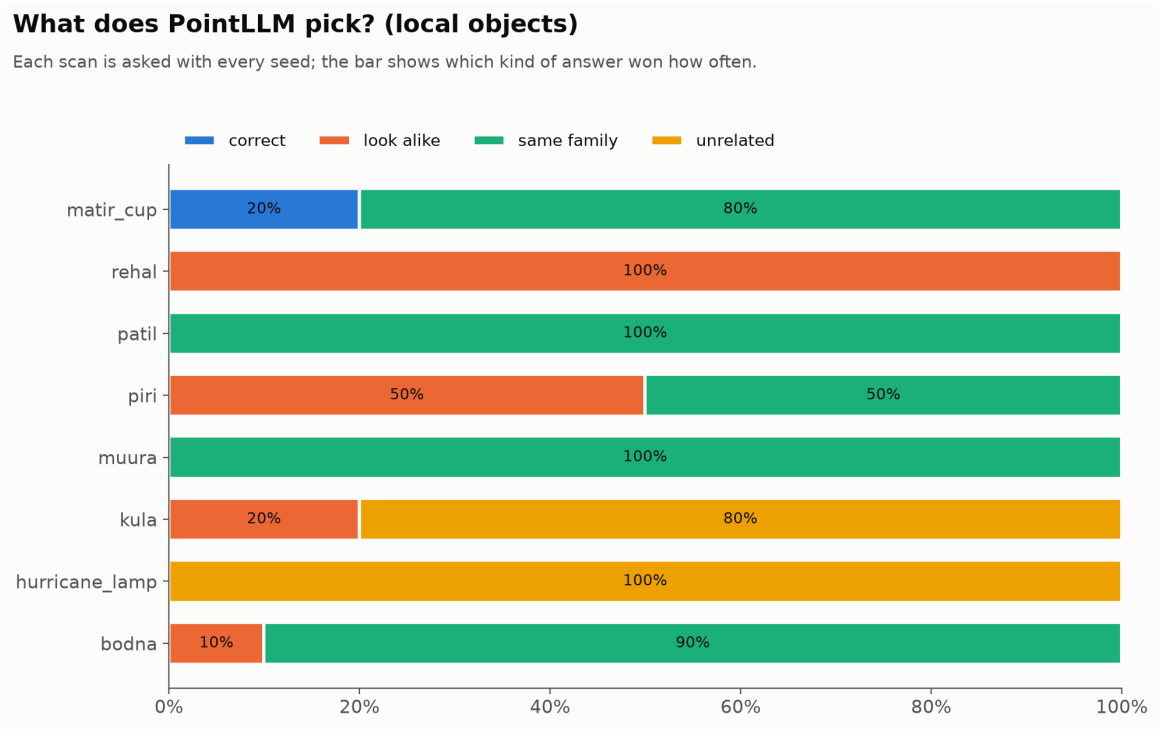


Figure 13. Local objects: almost always a familiar look-alike or a related object. The correct answer wins only in a minority of seeds for matir cup (20 %), and never as the majority pick.

Object	Correct answer	PointLLM picks
rehal	book stand	bench
patil	cooking pot	bowl
matir cup	clay cup	bowl (a near tie)
bodna	water jug	vase / teapot
piri	low stool	bench / box
muura, kula, hurricane lamp	cane stool, bamboo tray, lantern	chair (the most common word in training)

- **It reads the shape:** the blind inputs get different picks than the real scans, and for matir cup and Patil_2 the shape itself favours the correct answer once common words are discounted.
- **But it only has Western categories:** it files each local object under the nearest one it knows.
- **A mistake we caught:** our first scoring divided by answer length and favoured long answers (even random noise looked “correct”). We found it with the blind inputs, fixed it and re-scored the saved data.

Calibrated view. Common words like “chair” score high even for a ball or noise. Subtracting that baseline shows whether the *shape itself* points to the right answer:

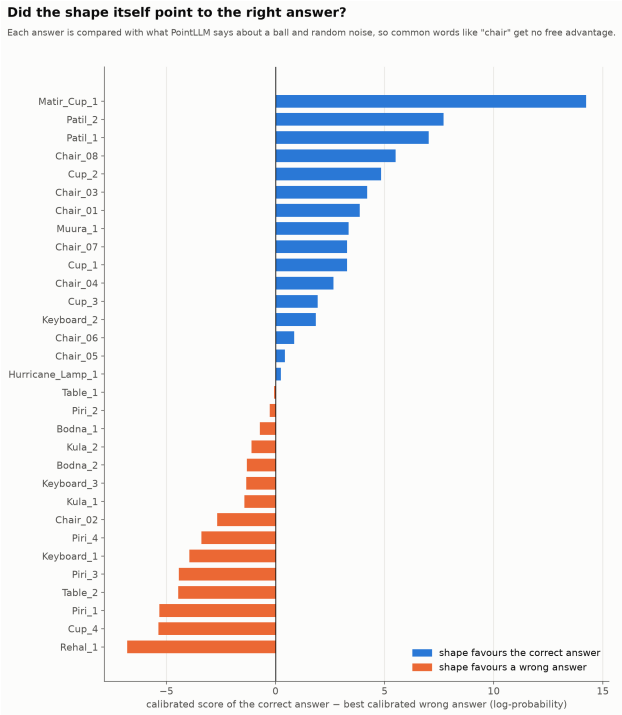


Figure 14. Run 1 (long answers). Blue = the shape favours the correct answer. Local: 5 of 14 (matir cup, patil ×2, muura, hurricane lamp). Rehal and piri clearly favour “bench”.

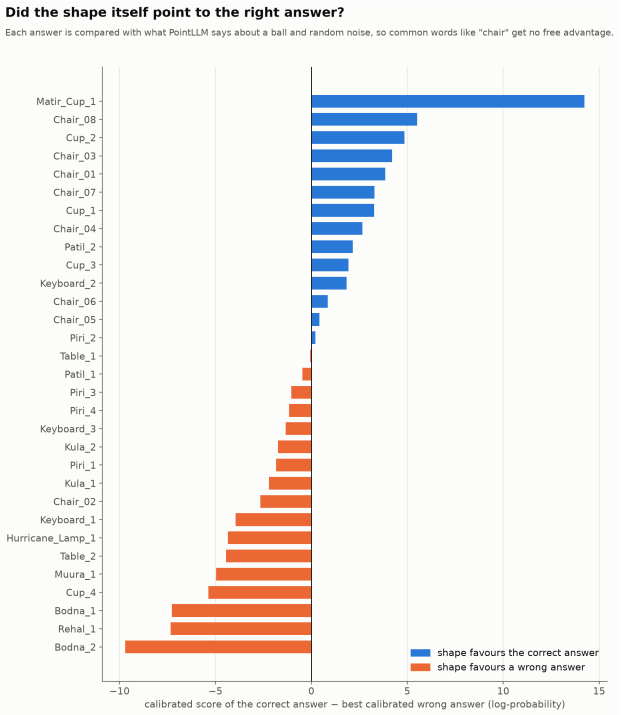


Figure 15. Run 2 (short answers, 1–2 words). Local: 3 of 14. Plain picks are identical to run 1 (still 0 of 14), so the result doesn't depend on how the answers are worded.

What it contributes: the main finding so far, with a method that separates “doesn't see the shape” from “sees it but has no name for it”.

2.7 CLIP-style model (MetaCLIP), first results

MetaCLIP is a 2D CLIP model: it scores how well a picture matches each text label. Shahedul rendered each 8192-point scan as depth-shaded pictures and tested 11 local scans with 4–6 labels per object. He also added the local name in English letters and in Bangla script. Three runs with the labels shuffled gave the same answers.

Object	Labels offered	CLIP's top pick	PointLLM's pick
bodna	kettle / teapot / cup / bottle	kettle / teapot	vase / teapot
patil	pot / bowl / saucer / basin	bowl	bowl
kula	basket / bowl / tray / plate	plate	chair
muura	basket / flower pot / bucket / stool	stool (99 %)	chair
piri	bench / chair / stool / table	stool	bench / box
rehal	book stand / chair / small table / bench	small table	bench
wooden bench	bench / table / stool / chair	bench	–

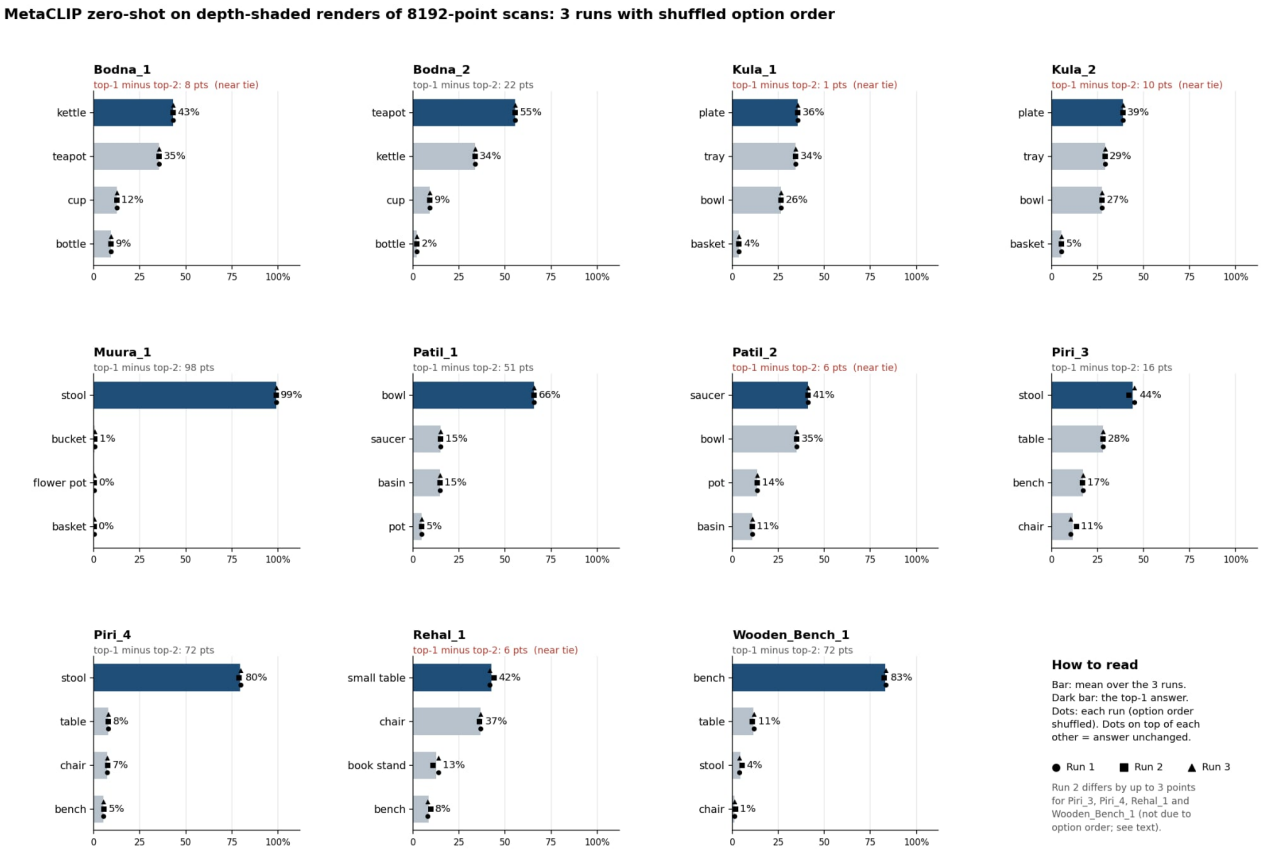


Figure 16. MetaCLIP on 11 local scans, 4 labels each, 3 runs with the label order shuffled (dots). Dark bar = top pick. Red subtitles mark near ties (top-1 minus top-2 under 11 points).

- Both models make the **same kind of mistake**: patil → bowl, bodna → teapot.
- CLIP is better on stool-like shapes and picks a good English **description** when it is offered (“folding book rest” for rehal, “floor stool” for piri).
- Local **names** (“bodna”, “rehal”, in English letters or Bangla script) score very low: the text side only knows English well.
- Not yet a fair comparison: the labels differ from PointLLM's options. Fixing that is week 1 of the plan.

What it contributes: a second model family, showing the problem isn't specific to PointLLM.

2.8 Literature: LiDAR vs photogrammetry on phones

Published tests say the two give similar overall shapes but differ on thin, shiny or small parts; iPhone LiDAR is about ±1 cm on objects above 10 cm (Luetzenburg et al. 2021). We use Scaniverse's mesh export throughout.

What it contributes: background for choosing a capture method for vehicles.

3. What is left: week by week

Week	Dates	Work	Output
1	2–8 Nov	Run CLIP with exactly PointLLM's answer options on all 14 local + 17 everyday scans. A teammate re-checks the scoring. Add answer options for the wooden bench.	Fair PointLLM vs CLIP table
2	9–15 Nov	Scan more local objects and more takes per object (goal: ≥ 3 scans each). Start local vehicles (rickshaw, thelagari, CNG) if access allows.	Bigger cleaned dataset
3	16–22 Nov	Clean the new scans; run the OOD check and multiple choice on them with both models. Rotation test (which way is “up” for the models).	Results on new objects
4	23–29 Nov	**Teaching pilot:** can a few examples per object teach the model? (nearest-neighbour or a small classifier on the models' shape fingerprints, tested on held-out scans)	Accuracy before vs after
5	30 Nov–6 Dec	Package the dataset (labels, metadata, capture protocol, README). Final analysis and figures.	Dataset release + final figures
6	7–13 Dec	Final report and presentation; rehearse; every member can explain every part.	Final presentation

4. What the final report will include

- **A small labelled dataset** of cleaned smartphone scans of Bangladeshi objects, plus the everyday reference set, with a capture and cleaning protocol others can follow.
- **A benchmark** for “near-OOD” objects: training-word check, shape-distance check, and multiple choice with blind inputs.
- **A comparison of two model families** (PointLLM vs CLIP-style) on the same questions.
- **A failure analysis:** which familiar object each local object is mistaken for, and why.
- **A teaching pilot:** whether a few examples are enough for a model to learn local objects.
- **Vehicles** (rickshaw, thelagari, CNG), if scanning access allows.
- **Reproducible notebooks** on free Google Colab, with all result tables in the repository.